IN-DETAIL: VA DISPLAYS

Bringing the best of both worlds, VA panels were made to compensate for the drawbacks of TN and IPS panels

It should be pretty obvious by now that LCD display panels primarily work by blocking light or allowing a fraction of light to pass through the coloured substrate. Back when different panel technologies were being invented, it was found that TN had fast response times but limited viewing angles, IPS had excellent colour reproduction but poor response times. So in 1996, Fujitsu developed the VA (Vertical Alignment) panel technology to offer the best of both worlds. Obviously, this was too good to be true as the first iteration of this panel technology had limited viewing angles, something the VA panel was supposed to improve upon. Hence, further improvements led to the inception of two distinct subtypes – MVA (Multi-domain Vertical Alignment) and PVA (Patterned Vertical Alignment).

Since then, VA panels have evolved and come quite a long way. They boast of improved viewing angles and colour reproduction compared to TN and response times that are much better than IPS panels. This is primarily due to the way VA panels operate. They're much better at blocking stray light from the backlight that might cause a shade of colour to appear different from what was intended. However, this comes at a price, VA isn't as